

Program : Diploma in Instrumentation Engineering	
Course Code : 5087	Course Title: Industrial Instruments Lab
Semester : 5	Credits: 1.5
Course Category: Program Core	
Periods per week : 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To provide hands on experience of the measurement techniques of various physical parameters.
- To equip to calibrate various instruments.

Course Prerequisites:

Topic	Course code	Course name	Semester
Theory of Transducers.		Sensors and transducer	3
Working of Transducers.		Sensors and Transducers lab	3
Temperature, Pressure, Level, flow measurements.		Process variable measurements	4
Humidity, Viscosity, Speed, Density, Acceleration, pH measurement.		Industrial instrumentation	5

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Plot the characteristics of temperature transducers	9	Applying
CO2	Calibrate thermometers and Bourdon tubes	2	Applying
CO3	Measure flow using orifice meter, venturi meter and Rotameter	9	Applying
CO4	Measure different physical parameters using various transducers	22	Applying
	Lab Test	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Plot the characteristics of temperature transducers		
M1.01	Plot the response curve of thermocouples with and without thermowell	2	Applying
M1.02	Plot the response curve of series combination of thermocouple.	1	Applying
M1.03	Plot the response curve of parallel combination of Thermocouple	1	Applying
M1.04	Plot the characteristics of Pt 100 R.T.D and determine the sensitivity.	1	Applying
M1.05	Determine the temperature coefficient of resistance of Pt 100 R.T.D	2	Applying
M1.06	Plot the characteristics of Thermistors	2	Applying
CO2	Calibrate thermometers and Bourdon tubes		
M2.01	Calibration of Bimetal thermometer	1	Applying
M2.02	Calibrate Bourdon tube Pressure gauge using dead weight Tester.	1	Applying
	Lab Test -I	1	
CO3	Measure flow using orifice meters, venturi meters and Rotameters		
M3.01	Measurement of flow by differential pressure method-Orifice meter and Venturimeter.	3	Applying
M3.02	Measurement of Flow by variable area flow meter –Rotameter.	2	Applying
M3.03	Determine Coefficient of Discharge of Venturimeter	2	Applying
M3.04	Determine Coefficient of Discharge of Orifice plate	2	Applying

CO4	Measure different physical parameters using various transducers.		
M4.01	Pressure measurement Using Well/ U-tube/ inclined tube manometers	1	Applying
M4.02	Pressure measurement Using Bourdon Tubes– C type / Spiral type iii. Capsules Bellows	1	Applying
M4.03	Pressure measurement Using Bellows	1	Applying
M4.04	Level measurement Using Capacitive transducers	1	Applying
M4.05	Level measurement using DP transmitter	1	Applying
M4.06	Determine Humidity Using psychrometer	1	Applying
M4.07	Determine Humidity Using hygrometer	1	Applying
M4.08	Determine Kinematic and Absolute Viscosity using Viscometer.	2	Applying
M4.09	Measure angular velocity using Stroboscope.	1	Applying
M4.10	Measure angular velocity using Tachometer	1	Applying
M4.11	Measure speed using AC and DC tacho generators	1	Applying
M4.12	To measure the force and torque using strain gauges.	2	Applying
M4.13	Measure Liquid Level Using Sight Glass and Float level sensor.	2	Applying
M4.14	Measure Liquid Level Using Conductive probes.	2	Applying
M4.15	Measure density using hydrometer.	1	Applying
M4.16	Measure the pH value of a solution using pH meter	1	Applying
M4.17	Plot the characteristics of acceleration/ vibration measurement	1	Applying
M4.18	To perform the operation of smoke/ gas detector	1	Applying
	Lab Test II	2	

Text / References:

T/R	Book Title /Author
R1	A K Sawney–Mechanical Measurements and Instrumentation Dhanpath Rai and sons .
R2	S. K. Singh – Industrial Instrumentation and Control-TMH.
R3	A K Sawney– Electrical Measurements and Instrumentation Dhanpath Rai and sons

Online Resources:

Sl.No	Website Link
1	https://www.youtube.com/watch?v=4knAR4HzRJs
2	www.me.umn.edu/courses/me4331/FILES/VolFlowMeasurements.ppt · Web view
3	https://www.slideshare.net/om_singh/flow-measurement
4	https://www.youtube.com/watch?v=XIIViaNITlw
5	www.emerson.com/.../measurement-instrumentation/flow-measurement
6	www.reference.com › Science › Physics › Motion & Mechanics
7	https://study.com/.../what-is-acceleration-definition-and-formula.html